The Geonovum OGC Checker

A linter for OGC standards

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Datum donderdag 9 oktober 2025

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Context

- Geospatial data standards are part of the Dutch ‘comply or explain’ list of National standards
- One of Geonovum’s tasks is to support implementation and adoption of these standards, e.g. through
  - Dutch profiles
  - Best practice documents
  - Examples
  - Implementation in open source tooling through e.g. testbeds
  - **Validation tooling**

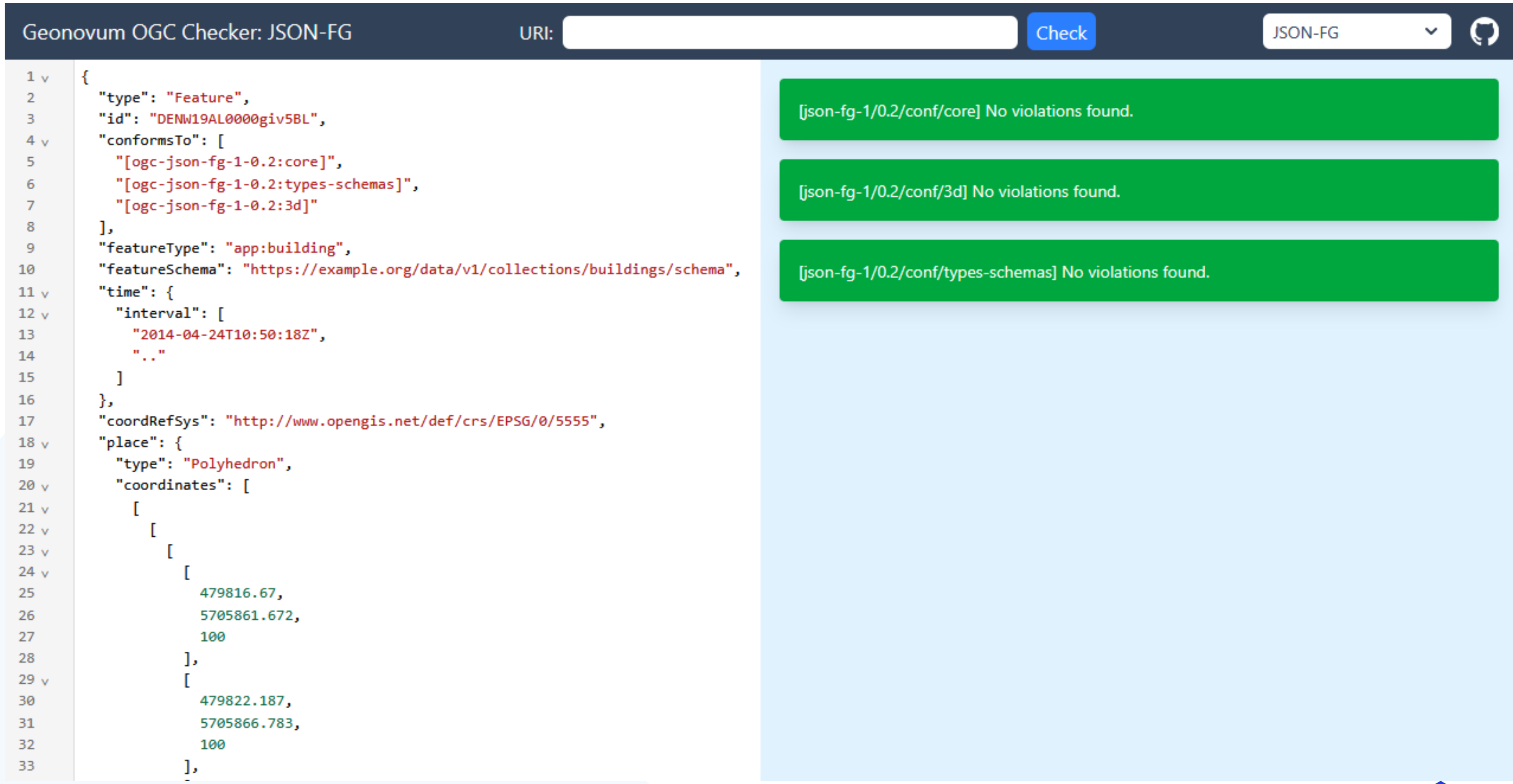
- At an OGC code sprint in 2023 we started to work on the “OGC checker”
  - A linter for OGC standards
  - First version created in 3 days, support for JSON-FG
  - Supports several OGC standards
  - Still under development

“We” = me and Joost Farla

The OGC checker

- Open source: https://github.com/Geonovum-labs/ogc-checker
- Runs as online validator at https://geonovum-labs.github.io/ogc-checker/
- As of now, supports validation of:

JSON-FG
OGC API Features part 1+2
OGC API Records
OGC API Processes



Checker for OGC API Features

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What it checks:

- JSON Syntax
- Conformance to OAS 3.0
- OGC API Features part 1 v1.0.1 validation rules
- OGC API Features part 2 v1.0.1 validation rules

Requirement	Testable	Tested	Remarks
`/req/core/root-op`	Yes	Yes	
`/req/core/root-success`	Yes	Yes	
`/req/core/conformance-op`	Yes	Yes	
`/req/core/conformance-success`	Yes	Yes	
`/req/core/fc-md-op`	Yes	Yes	
`/req/core/fc-md-success`	Yes	Yes	
`/req/core/sfc-md-op`	Yes	Yes	
`/req/core/sfc-md-success`	Yes	Yes	
`/req/core/fc-op`	Yes	Yes	
`/req/core/fc-response`	Yes	Yes	
`/req/core/fc-limit-definition`	Yes	Yes	
`/req/core/fc-bbox-definition`	Yes	Yes	
`/req/core/fc-time-definition`	Yes	Yes	
`/req/core/f-op`	Yes	Yes	
`/req/core/f-response`	Yes	Yes	
`/req/oas30/oas-definition-2`	Yes	Yes	

Requirement	Testable	Tested	Remarks
`/req/crs/crs-uri`	No	No	
`/req/crs/fc-md-crs-list`	Yes	Yes	
`/req/crs/fc-md-storageCrs`	No	No	
`/req/crs/fc-md-storageCrs-valid-value`	Yes	Yes	
`/req/crs/fc-md-crs-list-global`	No	No	
`/req/crs/fc-bbox-crs-definition`	Yes	Yes	
`/req/crs/fc-bbox-crs-valid-value`	Yes	Yes	Tests the presence of a 400 response.
`/req/crs/fc-bbox-crs-valid-defaultValue`	Yes	Yes	Covered by `/req/crs/fc-bbox-crs-definition`.
`/req/crs/fc-bbox-crs-action`	No	No	
`/req/crs/fc-crs-definition`	Yes	Yes	
`/req/crs/fc-crs-valid-value`	No	No	
`/req/crs/fc-crs-default-value`	Yes	Yes	Covered by `/req/crs/fc-crs-definition`.
`/req/crs/fc-crs-action`	No	No	
`/req/crs/geojson`	No	No	
`/req/crs/ogc-crs-header`	Yes	Yes	
`/req/crs/ogc-crs-header-value`	No	No	

Problem: profile based negotiation

Table 17 — API — Records MIME Types

Encoding		Media Type	Profile Identifier
Record	HTML	text/html	n/a
	JSON	application/json	ogc-record
	GeoJSON	application/geo+json	ogc-record
Catalog	HTML	text/html	n/a
	JSON	application/json	ogc-catalog
		application/ogc-catalog+json	n/a

- Content in an OGC API is identified by media type + profile identifier
- **This means that the direct correlation between media type and OAS content schema is lost**
- This means the linter cannot differentiate the content and therefore cannot check content against its specific schema

Sprint goals

- Have a discussion on profile based negotiation with OGC API SWG members
  - Adapt linter according to outcome

- Backlog: https://github.com/orgs/Geonovum-labs/projects/1

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Thank you!

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